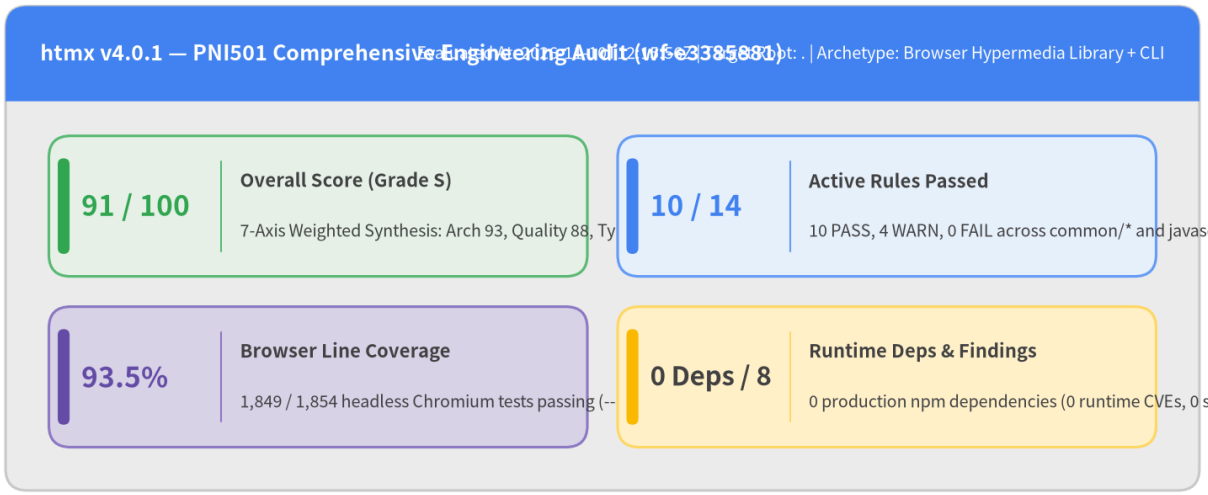
htmx v4.0.1 — PNI501 Software Health Diagnostic Report

Project: `htmx` (`v4.0.1`) | **Language**: `javascript` (`ES2022` + `TypeScript d.ts`) | **Workflow**: `pni501` (`wf-e3385881`)



PNI501 Software Health Scorecard for htmx v4.0.1 (Overall Score: 91/100, Grade S)

Audit Metadata & Scope Summary

Attribute	Value
Target Repository	`htmx` (`v4.0.1` , upstream tag `8d112c71`)
Primary Language & Runtime	JavaScript (`ES2022` classes & `#` private fields) + TypeScript Declarations (`src/htmx.d.ts`)
Runtime Profiles	Browser Hypermedia Library (`dist/htmx.js` , `dist/htmx.esm.js` , `dist/htmax.js`) & Node CLI (`src/scripts/upgrade-check.js`)
Workflow & Run ID	`pni501` (`wf-e3385881`) — Evaluated `2026-11-10T12:15:56+00:00`
Overall Score & Grade	**91 / 100** (`Grade S`)
Codebase Footprint	`18` source files (`5,521` non-blank LOC: `src/htmx.js` at `1,989` LOC + `17` extensions in `src/ext/*.js`)

Attribute	Value
Hermetic Browser Verification	`1,854` real-browser tests via `@web/test-runner` + Playwright Chromium (`1,849` passed, `1` timing race, `4` skipped)
Measured V8 Coverage	**93.5%** line coverage (`6,267 / 6,706` statements) and **92.3%** branch coverage
Companion Portals	User Guide • API Reference Catalog • Internal Architecture Guide

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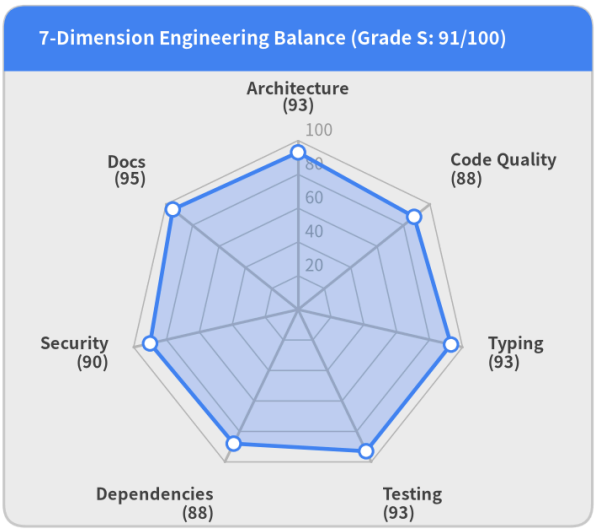
4. Audit Standards, Rule Compliance & Findings

5. Deep Dive: Code Quality, Testing & Runtime

6. Deep Dive: Security, Supply Chain & API Surface

1. Executive Summary

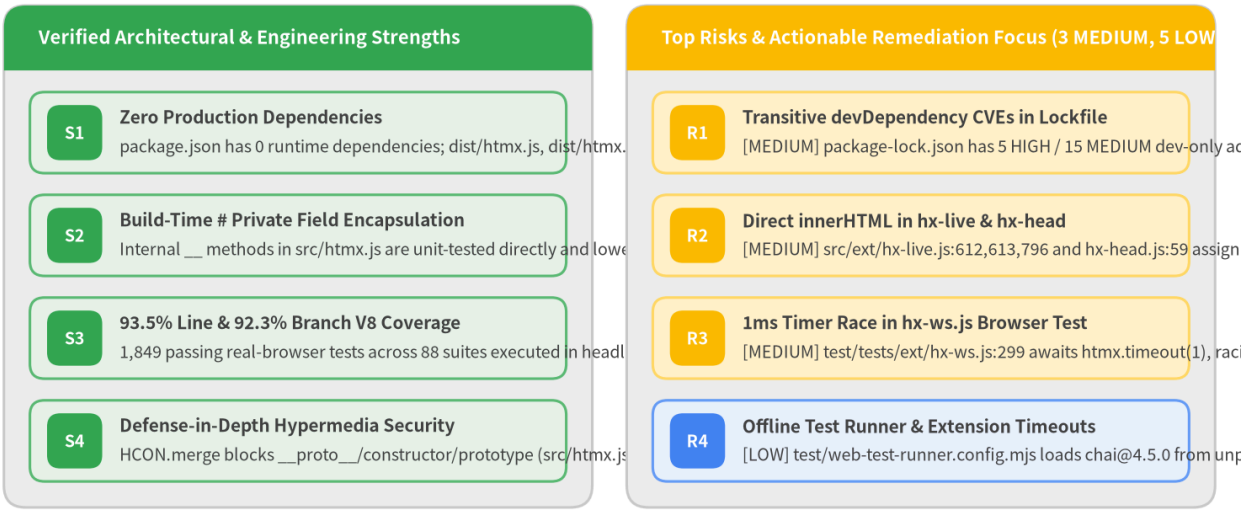
`htmx` `v4.0.1` achieves an overall **PNI501 Software Health Score of 91 / 100 (`Grade S`)**, passing `10` of `14` active engineering rules with `4` warnings and `0` failures. The ground-up v4 rewrite replaces htmx 2's legacy `XMLHttpRequest` closure with a zero-dependency ES2022 class engine (`Htmx` , `RequestQueue` , `HCON`) backed by native `fetch()` , `AbortController` , `ReadableStream` , built-in Idiomorph DOM morphing, and `17` acyclic official extensions.



Category Score & Audit Step Summary

Dimension	Score	Key Evidence
Architecture	93 (S)	0 cycles; 15-method #internalAPI
Code Quality	88 (A)	ES2022 classes; 4 loose != in morph
Typing	93 (S)	772-line src/htmx.d.ts; 28 events
Testing	93 (S)	1,849/1,854 browser tests; 93.5% cov
Dependencies	88 (A)	0 runtime deps; 5 HIGH dev CVEs
Security	90 (S)	Trusted Types + hx-csp; 0 secrets
Documentation	95 (S)	4 skill guides + 4 Diataxis portals

7-Dimension Engineering Health Radar & Category Score Breakdown for htmx v4.0.1



Headline Engineering Strengths vs. Top Prioritized Risks in htmx v4.0.1

Key Executive Takeaways

1. Seven-Axis Engineering Scorecard

Axis	Score	Grade	Rationale Summary
Architecture	`93`	S	Clean Microkernel + Event-Driven Extension Hook architecture: zero-dependency core (`src/htmx.js` , `1,989` non-blank LOC) with `17` acyclic IIFE extensions interacting solely through `htmx.registerExtension()` and a 15-method `#internalAPI` bridge.
Code Quality	`88`	A	Idiomatic ES2022 classes (`HCON` , `RequestQueue` , `Htmx`) using native `fetch()` , `AbortController` , `ReadableStream` , and constructable `CSSStyleSheet` ; minor deductions for `4` loose `!=` checks in morphing and `2` extension `fetch()` calls omitting `AbortSignal` .
Typing	`93`	S	Comprehensive `772` -line TypeScript contract (`src/htmx.d.ts`) typing `Htmx` , `HtmxConfig` (`36` options), `HtmxRequestCtx` , `HtmxEventMap` (`28` colon-separated events), and reactive `HxLive` proxies.
Testing	`93`	S	`1,854` real-browser test cases (`1,849` passed, `4` skipped, `1` timing race in `hx-ws.js:299`), achieving **93.5% line coverage** (`6,267 / 6,706` statements) and **92.3% branch coverage** in headless Chromium.
Dependencies	`88`	A	Zero production runtime dependencies (`dependencies: {}`) and a committed `package-lock.json` ; deducted for `5` HIGH and `15` MEDIUM transitive advisories in `devDependencies` (`cross-spawn` , `ip-address` , `basic-ftp` , `qs`).

Axis	Score	Grade	Rationale Summary
Security	`90`	**S**	Zero runtime supply-chain exposure, zero secrets, prototype pollution guard in `HCON.merge`, W3C Trusted Types (`#ttPolicy`), `hx-ignore` sandboxing, and eval-free `hx-csp.js`; `hx-live.js` and `hx-head.js` should route `.innerHTML` through `#ttPolicy`.
Documentation	`95`	**S**	Shipped developer/LLM skill guides (`src/skills/*.md`), `CHANGELOG.md`, `src/htmx.d.ts`, and a 4-portal bilingual documentation suite (`user_guide_src/`, `api_src/`, `architecture_src/`, `pni501_src/`).

2. Lowest-Scoring Audit Steps (Sorted Ascending)

1. `security_vulnerability_audit` (`88 / 100`, **Grade A**): Zero production dependency CVEs and zero secrets; warnings recorded for `22` transitive `devDependency` CVEs in `package-lock.json` (`5` HIGH, `15` MEDIUM, `2` LOW), `2` extensions (`hx-live.js`, `hx-head.js`) bypassing core's Trusted Types wrapper on `.innerHTML`, and missing `SECURITY.md` / Dependabot automation.
2. `javascript_ast_audit` (`89 / 100`, **Grade A**): `100%` import hygiene and zero circular dependencies across `18` modules (`5,521` LOC); minor warnings for `4` loose `!=` checks in `src/htmx.js:2044,2100,2127,2194` and `2` extension `fetch()` calls (`hx-preload.js:54`, `hx-download.js:33`) without `AbortSignal`.
3. `repo_structure_audit` (`90 / 100`, **Grade S**): Clean zero-dependency package structure and complete multi-format export map (`main`, `module`, `jsdelivr`, `unpkg`, `types`, `web-types`, `bin`); minor note to add `engines.node` and a static linter script in `package.json`.

2. Architecture & Source Code Structure

`htmx` `v4.0.1` implements a **Microkernel + Event-Driven Extension Hook Architecture**. The core hypermedia engine resides in a single self-contained module (`src/htmx.js` , `1,989` non-blank LOC) that has **zero outgoing imports**, while `17` official opt-in extensions (`src/ext/*.js` , `3,532` combined non-blank LOC) register lifecycle hooks (`htmx_before_request` , `htmx_after_swap` , etc.) via `htmx.registerExtension()` and receive a curated 15-method `#internalAPI` bridge in `init(api)` .

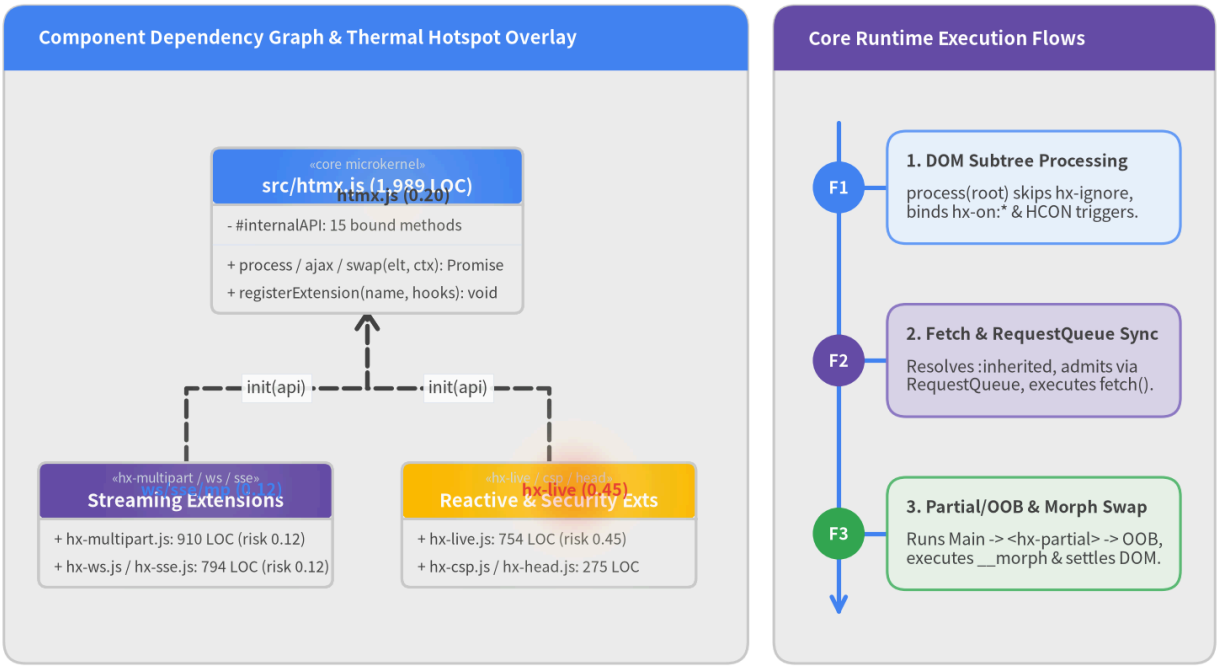
Repository & Source Directory Tree

- htmx (v4.0.1, 5,521 LOC)
  - **src/ (18 JS files + d.ts)**
    - htmx.js (1,989 LOC: HCON, RequestQueue, Htmx)
    - htmx.d.ts (772 LOC: Public TS Declarations)
    - **ext/ (17 Official IIFE Extensions)**
      - hx-multipart.js (910 LOC) & hx-live.js (754 LOC)
      - hx-ws.js (456 LOC) & hx-sse.js (338 LOC)
      - hx-history-cache.js (204) & hx-csp.js (140)
      - 11 UX/Compat Extensions (630 LOC total)
    - scripts/upgrade-check.{js,py} (v2->v4 CLI)
    - skills/*.md (4 Official Developer/AI Guides)
  - **dist/ (IIFE, ESM, Minified & htmax Bundles)**
  - **test/ (88 Suites, 1,854 Browser Tests)**

Source Module Inventory (18 JS Modules, 5,521 Non-Blank LOC)

Module Path	LOC	Architectural Role
src/htmx.js	1,989	Core Htmx, RequestQueue, HCON & Morph
src/ext/hx-multipart.js	910	Streaming multipart/mixed swapper
src/ext/hx-live.js	754	Reactive DOM proxy & q()/$() engine
src/ext/hx-ws.js	456	Bidirectional WebSocket transport
src/ext/hx-sse.js	338	SSE EventSource & Fetch stream client
src/ext/hx-history-cache.js	204	sessionStorage DOM snapshot cache
src/ext/hx-csp.js	140	Eval-free strict CSP AST interpreter
src/ext/hx-head.js	135	Document <head> merge & deduplication
src/ext/* (10 modules)	595	preload, compat, pending, upsert, etc.

Physical Repository Tree & Top Source Modules by Non-Blank Lines of Code (5,521 Total LOC)



Layered Microkernel Dependency Graph with Thermal Risk Overlay (HeatmapOverlay) & Execution Flows

Architectural Assessment & Module Inventory

1. Four-Layer Structural Separation

Layer	Repository Paths	Allowed Dependencies	Responsibility & Boundary Notes
Core Hypermedia Runtime & DOM Reconciler	`src/htmx.js`, `src/htmx.d.ts`	*(None — zero imports)*	Implements `HCON` relaxed configuration parser, `RequestQueue` concurrency coordinator, explicit `:inherited` attribute lookup, `fetch()` orchestration, `<hx-partial>` / OOB task extraction, and built-in Idiomorph DOM tree reconciliation.
Streaming & Transport Extensions	`src/ext/hx-sse.js`, `src/ext/hx-ws.js`, `src/ext/hx-multipart.js`, `src/ext/hx-preload.js`, `src/ext/hx-ptag.js`, `src/ext/hx-download.js`	`src/htmx.js`	Extends the Fetch and event lifecycle with Server-Sent Events (`EventSource` + `ReadableStream`), bidirectional WebSockets with exponential backoff, `multipart/mixed` streaming swaps, speculative prefetching, and `HX-PTag` conditional polling.

Layer	Repository Paths	Allowed Dependencies	Responsibility & Boundary Notes
Reactive Scripting, DOM & Security Extensions	`src/ext/hx-live.js`, `src/ext/hx-csp.js`, `src/ext/hx-history-cache.js`, `src/ext/hx-head.js`, `src/ext/hx-targets.js`, `src/ext/hx-upsert.js`, `src/ext/hx-pending.js`, `src/ext/hx-browser-indicator.js`, `src/ext/hx-prompt.js`, `src/ext/hx-alpine-compat.js`, `src/ext/htmx-2-compat.js`	`src/htmx.js`	Provides reactive DOM state proxies (`hx-live`), strict CSP AST expression evaluation without `eval` (`hx-csp`), `sessionStorage` history snapshot caching, `<head>` merging, and htmx 2 / Alpine.js compatibility bridges.
Migration & Tooling Scripts	`src/scripts/upgrade-check.js`, `src/scripts/upgrade-check.py`	*(Standalone CLI)*	Scans `.html`, `.js`, `.ts`, `.jinja`, `.erb`, and `.php` templates for htmx 2 to htmx 4 migration patterns (`hx-disable` -> `hx-ignore`, colon-separated events, explicit `:inherited`).

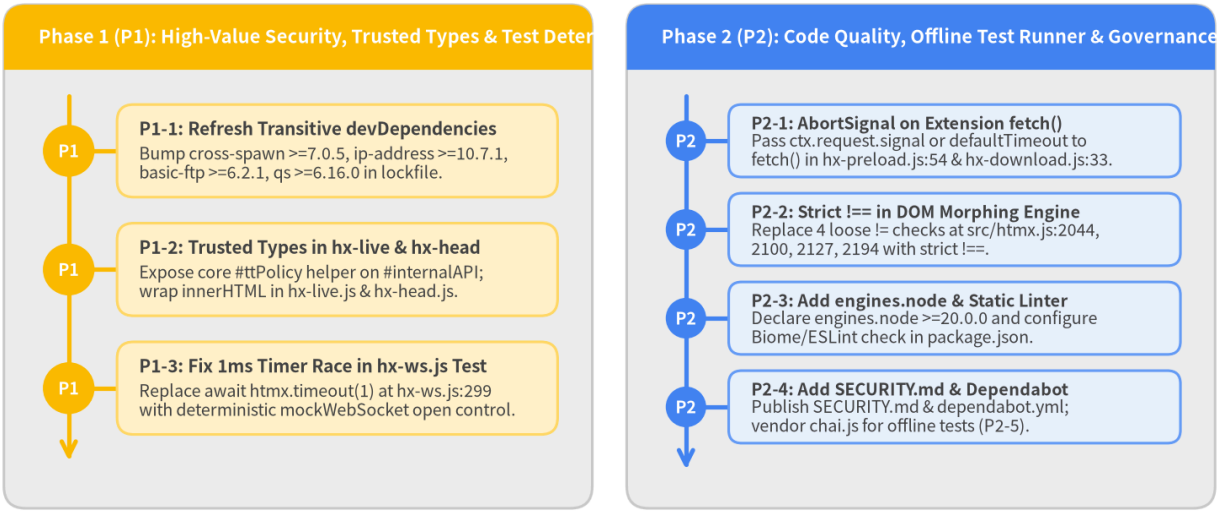
2. Complete Source Code Module Inventory & Thermal Hotspots

Module Path	Non-Blank LOC	Thermal Risk (`risk_score`)	Triggered Rules & Findings
`src/ext/hx-live.js`	`754`	`0.45`	`common/security_vulnerability` (`1 MEDIUM`: direct `.innerHTML` / `.outerHTML` at lines `612, 613, 796` bypassing `#ttPolicy`)
`src/htmx.js`	`1,989`	`0.20`	`javascript/code_quality` (`1 LOW`: `4` loose `!=` comparisons in `__morphChildren`, `__findBestMatch`, `__morphNode`)
`package-lock.json`	*(Lockfile)*	`0.15`	`common/security_vulnerability` (`1 MEDIUM`: `5 HIGH` / `15 MEDIUM` transitive `devDependency` CVEs)
`test/tests/ext/hx-ws.js`	*(Test Suite)*	`0.15`	`common/runtime_verification` (`1 MEDIUM`: `1ms` `htmx.timeout(1)` race at line `299`)
`src/ext/hx-multipart.js`	`910`	`0.12`	`0` findings (`81.9%` line coverage across `1,191` statements)
`src/ext/hx-ws.js`	`456`	`0.12`	`0` findings (`90.0%` line coverage across `588` statements)
`src/ext/hx-sse.js`	`338`	`0.12`	`0` findings (`95.0%+` line coverage)
`src/ext/hx-history-cache.js`	`204`	`0.12`	`0` findings (`sessionStorage` history snapshot cache)

Module Path	Non-Blank LOC	Thermal Risk (`risk_score`)	Triggered Rules & Findings
`src/ext/hx-csp.js`	`140`	`< 0.10`	`0` findings (`73.5%` line coverage; eval-free recursive AST interpreter)
`src/ext/hx-head.js`	`135`	`< 0.10`	Grouped under `hx-live.js:612` Trusted Types recommendation (`htmlDoc.innerHTML` at line `59`)
`src/ext/hx-preload.js`	`88`	`< 0.10`	`common/security` (`1 LOW` : `fetch()` at line `54` omits `AbortSignal` timeout)
`src/ext/hx-alpine-compat.js`	`84`	`< 0.10`	`0` findings (`96.4%` line coverage)
`src/ext/htmx-2-compat.js`	`83`	`< 0.10`	`0` findings (`81.7%` line coverage)
`src/ext/hx-pending.js`	`82`	`< 0.10`	`0` findings (`100%` line coverage)
`src/ext/hx-upsert.js`	`70`	`< 0.10`	`0` findings (`100%` line coverage)
`src/ext/hx-browser-indicator.js`	`69`	`< 0.10`	`0` findings (`100%` line coverage)
`src/ext/hx-download.js`	`60`	`< 0.10`	Grouped under `hx-preload.js:54` (`fetch()` at line `33` omits `AbortSignal`)
`src/ext/hx-targets.js`	`27`	`< 0.10`	`0` findings (`100%` line coverage)
`src/ext/hx-ptag.js`	`18`	`< 0.10`	`0` findings (`100%` line coverage)
`src/ext/hx-prompt.js`	`14`	`< 0.10`	`0` findings (`100%` line coverage)

3. Recommended Remediation Roadmap

The PNI501 workflow identified **8 prioritized engineering action items** (`0` `P0` blockers, `3` `P1` high-value improvements, and `5` `P2` hardening & polish enhancements). Because `htmx` `v4.0.1` already achieves `Grade S` (`91 / 100`) with zero production dependencies, this roadmap focuses on dev-lockfile CVE hygiene, Trusted Types consistency across extensions, and 100% deterministic offline browser testing.



Phased Remediation Roadmap (P1 High-Value Improvements -> P2 Hardening & Offline CI Polish)

Prioritized Action Plan Traceability Matrix (P1 -> P2, 8 Total Items)

Prio	Action Item Title	Target Files	Related Rules / CVEs	Expected Impact
P1	Refresh devDependencies in lockfile	package-lock.json	CVE-2024-21538, +3	Resolves 5 HIGH dev CVEs
P1	Route innerHTML via Trusted Types	src/ext/hx-live.js, +2	CWE-79, security_vuln	Full strict CSP compatibility
P1	Fix 1ms timer race in hx-ws test	test/tests/ext/hx-ws.js	runtime_verification	100% deterministic CI runs
P2	Attach AbortSignal to ext fetch()	src/ext/hx-preload.js	common/security	Prevents hung prefetch I/O
P2	Use strict !== in DOM morphing	src/htmx.js	javascript/code_quality	Eliminates coercive != checks
P2	Declare engines.node & linter	package.json	javascript/dependencies	Catches syntax issues early
P2	Add SECURITY.md & Dependabot	package.json	common/supply_chain	OpenSSF Scorecard compliance
P2	Vendor chai.js for offline WTR	test/web-test-runner...	runtime_verification	Out-of-box --network=none

Action Plan Traceability Matrix (Priority, Target Files, Related Rules/CVEs, and Expected Impact)

Detailed Action Items (`P1` -> `P2`)

1. `[P1]` **Refresh transitive** `devDependencies` **in** `package-lock.json` **to resolve 5 HIGH CVEs** (`dependencies`)
   - **Target Files**: `package-lock.json`, `package.json`
   - **Related Rules / CVEs**: `common/security_vulnerability`, `CVE-2024-21538`, `CVE-2026-69192`, `CVE-2026-102990`, `CVE-2026-82417`
   - **Action**: Update `package-lock.json` via `npm audit fix` or `package.json` `overrides` to bump `cross-spawn` to `>=7.0.5`, `ip-address` to `>=10.7.1`, `basic-ftp` to `>=6.2.1`, and `qs` to `>=6.16.0`.
   - **Expected Impact**: Eliminates all fixable HIGH and MEDIUM CVEs in the development and test toolchain lockfile.
2. `[P1]` **Route** `innerHTML` / `outerHTML` **assignments in** `hx-live.js` **and** `hx-head.js` **through core Trusted Types policy** (`security`)
   - **Target Files**: `src/ext/hx-live.js`, `src/ext/hx-head.js`, `src/htmx.js`
   - **Related Rules / CVEs**: `common/security_vulnerability`, `CWE-79`

- **Action**: Expose a Trusted Types helper on `#internalAPI` in `src/htmx.js` and use it in `src/ext/hx-live.js:612,613,796` and `src/ext/hx-head.js:59` before assigning to `.innerHTML` or `.outerHTML`.
- **Expected Impact**: Ensures full compatibility with `require-trusted-types-for 'script'` CSP policies across all official extensions.

3. `[P1]` **Eliminate 1ms timer race in `hx-ws.js` message queue unit test** (`testing`)

   - **Target Files**: `test/tests/ext/hx-ws.js`
   - **Related Rules**: `common/runtime_verification`, `javascript/testing`
   - **Action**: In `test/tests/ext/hx-ws.js:293-310`, trigger `mockWebSocket` open deterministically (or increase mock auto-open latency) after asserting `connection.queue.length === 1` instead of relying on `await htmx.timeout(1)`.
   - **Expected Impact**: Makes the 1,854-test Playwright browser suite 100% deterministic under high parallel concurrency.

4. `[P2]` **Attach `AbortSignal` timeout to extension `fetch()` calls in `hx-preload` and `hx-download`** (`code_quality`)

   - **Target Files**: `src/ext/hx-preload.js`, `src/ext/hx-download.js`
   - **Related Rules**: `common/security`, `javascript/code_quality`
   - **Action**: Propagate `ctx.request.signal` or `AbortSignal.timeout(htmx.config.defaultTimeout)` into the `fetch()` options in `src/ext/hx-preload.js:54` and `src/ext/hx-download.js:33`.
   - **Expected Impact**: Prevents hung background prefetch or download requests from leaking browser connections.

5. `[P2]` **Standardize strict equality (`!==`) in DOM morphing routines** (`code_quality`)

   - **Target Files**: `src/htmx.js`
   - **Related Rules**: `javascript/code_quality`
   - **Action**: Replace loose `!=` checks at lines `2044`, `2100`, `2127`, and `2194` of `src/htmx.js` with strict `!==` comparisons.
   - **Expected Impact**: Eliminates coercive equality checks in hot DOM tree reconciliation loops.

6. `[P2]` **Declare `engines.node` and static lint script in `package.json`** (`dependencies`)

- **Target Files**: `package.json`
- **Related Rules**: `javascript/dependencies`
- **Action**: Add an `engines` field specifying `Node >=20` and configure Biome or ESLint in `devDependencies` for static syntax/style verification.
- **Expected Impact**: Enforces consistent Node runtime expectations and catches syntax regressions prior to browser test runs.

7. `[P2]` **Add** `SECURITY.md` **and Dependabot configuration for automated supply-chain hygiene** (`security`)

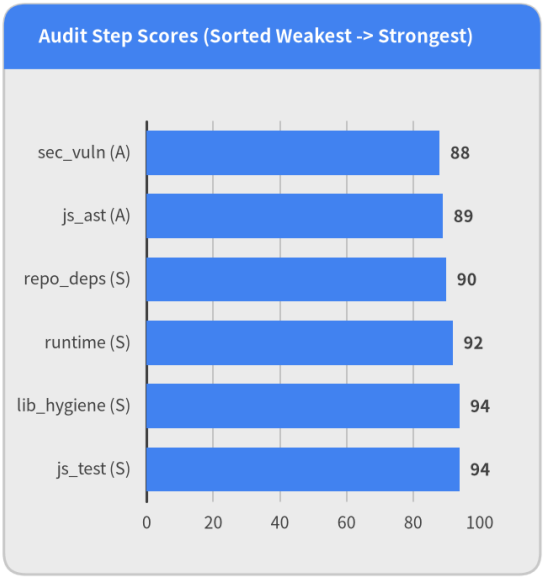
   - **Target Files**: `package.json`
   - **Related Rules**: `common/supply_chain`
   - **Action**: Create `SECURITY.md` with vulnerability reporting instructions and `.github/dependabot.yml` for automated `npm` lockfile PRs.
   - **Expected Impact**: Satisfies OpenSSF Scorecard supply-chain governance checks and prevents `devDependency` drift.

8. `[P2]` **Vendor or declare** `chai` **in** `devDependencies` **for offline test execution** (`testing`)

   - **Target Files**: `test/web-test-runner.config.mjs` , `package.json`
   - **Related Rules**: `common/runtime_verification` , `javascript/dependencies`
   - **Action**: Vendor `chai.js` under `test/lib/chai.js` (or add `chai` to `devDependencies`) and replace the `https://unpkg.com/chai@4.5.0/chai.js` script URL in `test/web-test-runner.config.mjs` .
   - **Expected Impact**: Allows `npm test` to execute in air-gapped and `--network=none` CI environments without external CDN access.

4. Audit Standards, Rule Compliance & Findings

The PNI501 workflow evaluated `htmx` `v4.0.1` across **6 executed audit steps** (`security_vulnerability_audit`, `javascript_ast_audit`, `repo_structure_audit`, `container_runtime_audit`, `library_hygiene_audit`, and `javascript_test_audit`) and **14 active engineering rules**. Overall rule compliance stands at **10 PASS, 4 WARN, and 0 FAIL**, with **8 raw findings** consolidated into **7 unique repository locations**.



Rule Compliance Matrix (WARN -> PASS, 14 Rules)

Status	Rule ID	Key Evidence
WARN	common/runtime_verification	1,849/1,854 pass; 93.5% cov
WARN	common/security_vulnerability	0 prod CVEs; 5 HIGH dev CVEs
WARN	common/supply_chain	Lockfile OK; no SECURITY.md
WARN	javascript/code_quality	4 loose !=; 2 fetch timeouts
PASS	common/architecture	0 cycles; 15-method API
PASS	common/library_guidelines	12 public + 86 # private
PASS	common/production_readiness	60s AbortController timeout
PASS	javascript/testing	88 suites; fetch-mock
PASS	javascript/typing (+5 more)	772-line src/htmx.d.ts

Executed Audit Step Scores (Weakest -> Strongest) & Rule Compliance Summary

Consolidated Findings by Location (CRITICAL -> LOW, 7 Locations)

Sev	Location (file:line)	Rule ID(s)	Deterministic Evidence	Recommended Fix
MED	package-lock.json:1	common/ security_vulnerability	5 HIGH, 15 MED dev CVEs	npm audit fix / overrides
MED	src/ext/hx-live.js:612	common/ security_vulnerability	Direct innerHTML/outerHTML	Wrap via #ttPolicy helper
MED	test/tests/ext/hx-ws.js:299	common/ runtime_verification	await htmx.timeout(1) race	Deterministic mock open
LOW	package.json:1	javascript/dependencies, common/supply_chain	No linter/SECURITY.md	Add engines, lint, policy
LOW	src/ext/hx-preload.js:54	common/security	fetch() without AbortSignal	Pass ctx.request.signal
LOW	src/htmx.js:2044	javascript/code_quality	4 loose != in morph engine	Use strict !== comparison
LOW	test/web-test-runner...:20	common/ runtime_verification	unpkg CDN chai@4.5.0 script	Vendor chai.js locally

Consolidated Location Findings Evidence Table (7 Unique Locations from 8 Raw Findings)

Full Rule Compliance & Findings Reference

1. Executed Audit Step Scores (Weakest -> Strongest)

Audit Step ID	Score & Grade	Verdict Counts	Step Synthesis
`security_vulnerability_audit`	**88 / 100 (`A`)**	`pass=0, warn=2, fail=0`	Production bundles (`dist/htmx.js` , `dist/htmx.esm.js` , `dist/htmax.js`) have `0` runtime dependencies, `0` leaked secrets, and `0` container misconfigurations. `package-lock.json` contains `22` `devDependency` advisories (`5` HIGH, `15` MEDIUM, `2` LOW), and `hx-live.js` / `hx-head.js` assign `.innerHTML` directly without core's `#ttPolicy` .
`javascript_ast_audit`	**89 / 100 (`A`)**	`pass=5, warn=1, fail=0`	`18` source modules (`5,521` non-blank LOC) with `0` circular dependencies and a `772` -line TypeScript declaration file (`src/htmx.d.ts`). Warnings note `4` loose `!=` checks in morphing and `2` extension `fetch()` calls omitting `AbortSignal` .
`repo_structure_audit`	**90 / 100 (`S`)**	`pass=5, warn=0, fail=0`	Zero runtime dependencies, `6` pinned `devDependencies` in `package-lock.json` , dual licensing (`LICENSE-HTMX` BSD-0-Clause + `LICENSE` Apache-2.0), and comprehensive multi-target exports in `package.json` .

Audit Step ID	Score & Grade	Verdict Counts	Step Synthesis
`container_runtime_audit`	**92 / 100 (`S`)**	`pass=1, warn=1, fail=0`	Hermetic Node 22 + Playwright Chromium image built in `127.33s` ; `1,854` real-browser tests executed with `--network=none` in `61.33s` (`1,849` passed, `4` skipped, `1` timer race at `hx-ws.js:299`), achieving **93.5% line coverage** and **92.3% branch coverage**.
`library_hygiene_audit`	**94 / 100 (`S`)**	`pass=3, warn=0, fail=0`	Strictly bounded public API (`12` public methods on `htmx` , `86` internal methods lowered via `sed` to ES2022 `#` private fields in `dist/`), silent-by-default logging (`logAll = false`), and multi-format bundles.
`javascript_test_audit`	**94 / 100 (`S`)**	`pass=1, warn=0, fail=0`	`88` browser test suites (`44` unit, `23` attributes, `6` end2end, `15` ext) with `1,849` Mocha/Chai `it()` blocks, deterministic `fetch-mock.js` isolation, and per-test DOM cleanup.

2. Complete Rule Compliance Matrix (`FAIL -> WARN -> PASS`)

Status	Rule ID	Source Action	Deterministic Evidence & Summary
WARN	`common/runtime_verification`	`runtime_verification`	`harness_build=passed (127.33s)` ; offline tests `total=1854, passed=1849, failed=1` (`hx-ws.js:299` `1ms` timer race), `skipped=4` ; V8 coverage `line=93.5%, branch=92.3%` (`6,267/6,706` statements); `smoke_check=passed` .
WARN	`common/security_vulnerability`	`security_vulnerability_audit`	`0` production runtime dependencies in `package.json` ; `0` secrets; `0` misconfigs; `22` `devDependencies` CVEs in `package-lock.json` (`5` HIGH: `CVE-2024-21538` , `CVE-2026-69192` , `CVE-2026-102990` , `CVE-2026-19693` , `CVE-2026-56876`); direct `innerHTML` assignments in `src/ext/hx-head.js:59` and `src/ext/hx-live.js:612,613,796` bypass core Trusted Types policy.
WARN	`common/supply_chain`	`security_vulnerability_audit`	`package-lock.json` committed (`0` unpinned specs); `LICENSE` present; `SECURITY.md=False` ; `dependency_automation=False` .
WARN	`javascript/code_quality`	`ast_and_quality`	`4` loose equality (`!=`) checks in `src/htmx.js:2044,2100,2127,2194` ; `6` intentional progressive-fallback `catch {}` blocks in `src/htmx.js` ; `2` `fetch()` calls without `AbortSignal` timeout in `src/ext/hx-download.js:33` and `src/ext/hx-preload.js:54` .
PASS	`common/architecture`	`ast_and_quality`	`0` circular dependencies across `18` modules; clean hub-and-spoke extension architecture via `htmx.registerExtension()` and `#internalAPI` .
PASS	`common/documentation`	`repo_and_deps`	Root `README.md` , `CHANGELOG.md` , `4` official skill guides in `src/skills/*.md` , `772` -line JSDoc-annotated `src/htmx.d.ts` , and `4` Diataxis portals.
PASS	`common/git_workflow`	`repo_and_deps`	`.gitignore` (`230` lines) excludes `node_modules/` , `coverage/` , `.drawlib/` , and `.pni/cache/` while tracking upstream `dist/` and lockfiles.

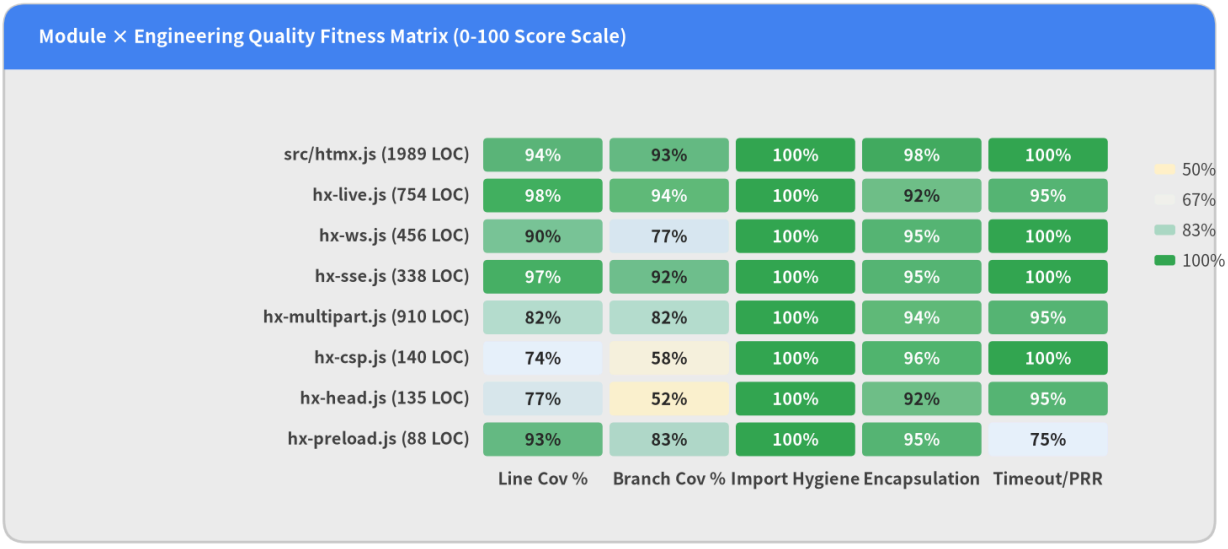
Status	Rule ID	Source Action	Deterministic Evidence & Summary
PASS	`common/library_guidelines`	`library_hygiene`	`package.json` exports `main`, `module`, `jsdelivr`, `unpkg`, `types`, and `web-types`; build scripts lower all `__` internal methods to `#` private fields; `0` runtime dependencies.
PASS	`common/maintainability`	`ast_and_quality`	`5,521` non-blank LOC across `18` files (`src/htmx.js` `1,989` LOC, `hx-multipart.js` `910` LOC, `hx-live.js` `754` LOC); `98` well-factored methods in `Htmx` class.
PASS	`common/production_readiness`	`runtime_verification`	Core requests enforce `defaultTimeout=60000ms` via `AbortController` (`__initTimeout` in `src/htmx.js:679`); `hx-ws.js` and `hx-sse.js` implement exponential backoff with jitter.
PASS	`common/security`	`repo_and_deps`	`0` committed secret files; `HCON.merge` explicitly blocks `__proto__`, `constructor`, and `prototype` (`src/htmx.js:75`); `#ttPolicy` Trusted Types hook and `hx-csp.js` extension.
PASS	`javascript/dependencies`	`repo_and_deps`	`0` runtime dependencies; all `6` build/test packages isolated in `devDependencies`.
PASS	`javascript/testing`	`test_health`	`88` test files across `test/tests/{unit,attributes,end2end,ext}/*.js` (`1,849` `it()` blocks); deterministic HTTP mocking via `test/lib/fetch-mock.js`.
PASS	`javascript/typing`	`ast_and_quality`	Companion `src/htmx.d.ts` (`772` lines) provides full TypeScript contracts for `Htmx`, `HtmxConfig`, `HtmxRequestCtx`, `HtmxEventMap`, and `HxLive`.

3. Configured Design Waivers (`pni.toml` `[[opt_out]]`)

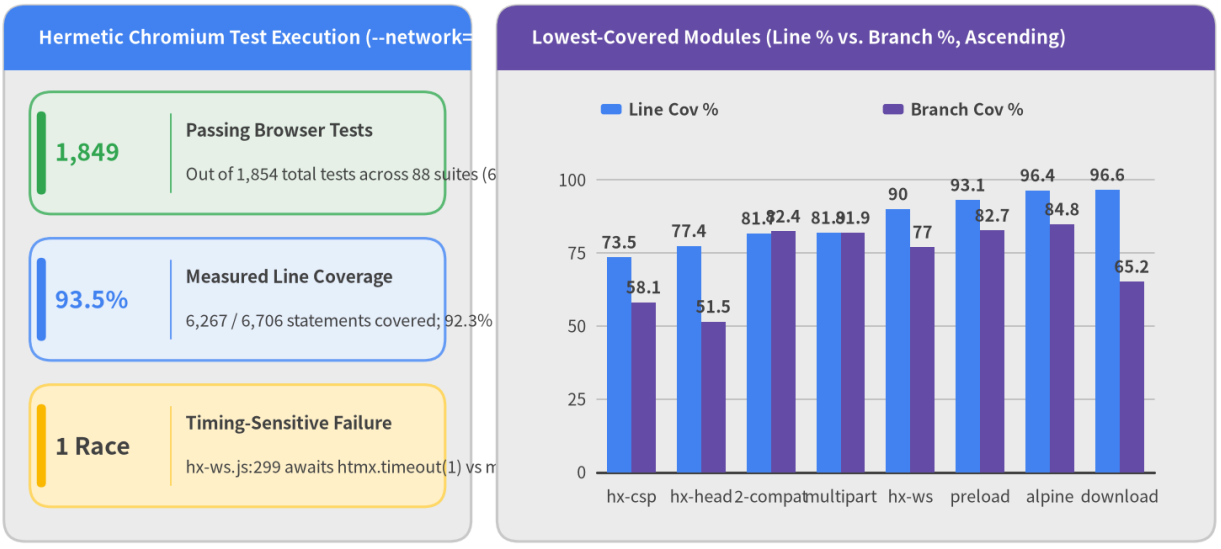
- `DS-0026` (scope: `.pni/dockerfiles/**`, matched: `0`): Hermetic batch test container exits upon `@web/test-runner` completion; daemon `HEALTHCHECK` is not applicable.

5. Deep Dive: Code Quality, Testing & Runtime

This chapter examines static JavaScript AST quality metrics (`5,521` non-blank LOC across `18` source modules), SIG maintainability indicators, 12-Factor operational readiness, and the hermetic Playwright Chromium browser test execution (`1,854` test cases, **93.5% line coverage**, **92.3% branch coverage**).



Module × Quality Dimension Fitness Matrix (HeatmapChart across Core and Key Extensions)



Hermetic Browser Test Execution Summary & Lowest-Covered Modules (Line vs. Branch Coverage %)

Static AST Quality & Containerized Runtime Analysis

1. Static AST Quality & SIG Maintainability Metrics

Metric Category	Deterministic Value	Technical Interpretation
Codebase Size	`18` files / `5,521` LOC	`src/htmx.js` (`1,989` non-blank LOC) + `17` extensions in `src/ext/*.js` (`3,532` non-blank LOC).
Circular Dependencies	`0`	Strictly acyclic hub-and-spoke topology (`src/htmx.js` has `0` outgoing imports).
Global State Usages	`1` (`src/htmx.js:1`)	Top-level `var htmx = (() => { ... })()` singleton required for `<script src="/htmx.js">` browser globals.
Loose Inequality (`!=`)	`4` (`src/htmx.js:2044..2194`)	Located in `__morphChildren`, `__findBestMatch`, and `__morphNode`; recommended to standardize on `!==`.
Empty `catch {}` Blocks	`6` (`src/htmx.js:40..2315`)	Deliberate progressive feature fallbacks (`HCON.parse` JSON probe, `CSSStyleSheet` indicator injection, `focus()`, `Element.moveBefore()`, View Transitions).
Missing `AbortSignal` Timeouts	`2` (`hx-preload.js:54`, `hx-download.js:33`)	Core `Htmx.__initTimeout` (`src/htmx.js:679`) enforces a `60,000ms` `AbortController` timeout on all primary requests; `hx-preload` and `hx-download` invoke `fetch()` directly without attaching `AbortSignal`.

Metric Category	Deterministic Value	Technical Interpretation
Console Logging	`2` (`src/htmx.js:1522`, `htmx-2-compat.js:30`)	`src/htmx.js:1522` is strictly gated behind `if (this.config.logAll)`; `htmx-2-compat.js:30` emits deliberate developer migration warnings.
Assertion-Free Test Candidates	`47` (all verified)	All `47` AST-flagged test blocks delegate assertions to shared helpers in `test/lib/helpers.js` (`assertTextContentIs`, `assertInnerHtmlIs`).

2. Hermetic Container Runtime & Browser Test Verification

- **Container Harness Build**: Built `.pni/dockerfiles/root/Dockerfile` (`node:22-bookworm` + Playwright Chromium + local `chai@4.5.0` cache) in `127.33s` (`status=passed`).
- **Offline Browser Test Suite (`--network=none`)**: Executed `1,854` browser tests across `88` suites in `61.33s`:
  - **Passed**: `1,849` (`99.73%` of active tests)
  - **Skipped**: `4`
  - **Failed**: `1` timing-sensitive race in `test/tests/ext/hx-ws.js:299`:

    ```
    [FAILURE] JEOsiBLigWWJDrt4YZ0H1 > hx-ws WebSocket extension > Message
    Sending.queues a message until the initial connection opens:
    AssertionError: expected +0 to equal 1
    ```

    Root Cause: Line `299` executes `await htmx.timeout(1)` before asserting `expect(connection.queue.length).to.equal(1)`. Under parallel multi-tab Chromium execution, the `1ms` timer can fire after `mockWebSocket`'s auto-open callback has already flushed the queue (`length === 0`).
- **Startup Smoke Check (`library+cli`)**: Verified both `dist/htmx.js` and `node src/scripts/upgrade-check.js --help` (`status=passed`, exit code `0`).

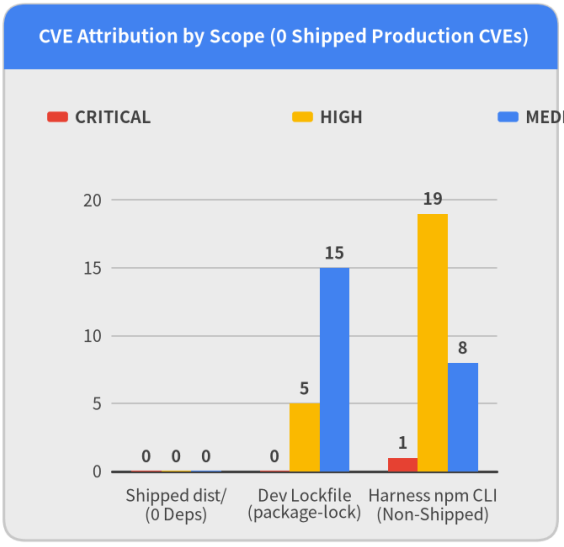
3. Lowest-Covered Modules (Measured V8 Coverage, Ascending)

Module Path	Line Coverage	Branch Coverage	Statements Covered	Notes
`src/ext/hx-csp.js`	`73.5%`	`58.1%`	`161 / 219`	Uncovered branches correspond to defensive AST operator/syntax error paths.

Module Path	Line Coverage	Branch Coverage	Statements Covered	Notes
`src/ext/hx-head.js`	`77.4%`	`51.5%`	`147 / 190`	Uncovered branches handle edge-case `<head>` script/link re-evaluation fallbacks.
`src/ext/htmx-2-compat.js`	`81.7%`	`82.4%`	`76 / 93`	Legacy htmx 2 event/attribute translation shims.
`src/ext/hx-multipart.js`	`81.9%`	`81.9%`	`975 / 1,191`	Streaming boundary chunk split and malformed header edge cases.
`src/ext/hx-ws.js`	`90.0%`	`77.0%`	`529 / 588`	WebSocket reconnect backoff and close-code branches.
`src/ext/hx-preload.js`	`93.1%`	`82.7%`	`95 / 102`	Speculative mouseover/mousedown prefetch triggers.
`src/htmx.js` *(Core Engine)*	**94.3%**	**93.1%**	`2,914 / 3,090`	Core `HCON`, `RequestQueue`, `Htmx`, `fetch()`, and Idiomorph engine.
`src/ext/hx-alpine-compat.js`	`96.4%`	`84.8%`	`106 / 110`	Alpine.js morph/swap state synchronization bridge.
`src/ext/hx-download.js`	`96.6%`	`65.2%`	`85 / 88`	Blob download and `Content-Disposition` filename extraction.
`src/ext/hx-live.js`	**97.6%**	**94.0%**	`1,042 / 1,068`	Reactive DOM state proxies and `q()` / `$()` query wrappers.

6. Deep Dive: Security, Supply Chain & API Surface

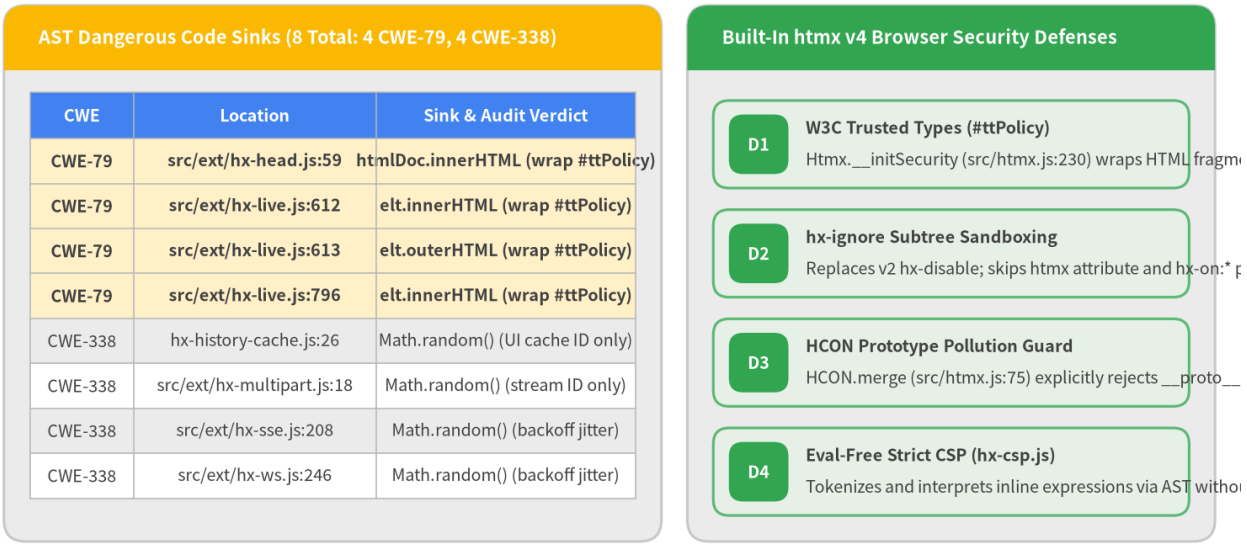
`htmx` `v4.0.1` ships **zero production runtime dependencies** (`dependencies: {}` in `package.json`), meaning end-user browsers loading `dist/htmx.js` , `dist/htmx.esm.js` , or `dist/htmax.js` inherit **zero third-party npm supply-chain vulnerabilities**. This chapter dissects the Trivy container/lockfile scan results, CWE-tagged DOM sinks, and built-in hypermedia security boundaries.



Key Advisories in Dev Lockfile & Harness Image

Sev	Scope	Package @ Version	Fixed Version
HIGH	dev_lockfile	cross-spawn@7.0.3	7.0.5
HIGH	dev_lockfile	ip-address@9.0.5	10.3.1
HIGH	dev_lockfile	basic-ftp@5.3.1	6.2.1
MED	dev_lockfile	qs@6.15.3	6.16.0
MED	dev_lockfile	postcss-sel...@6.1.0	6.1.5 / 7.1.6
CRIT	harness_npm	tar@6.2.1 (global npm)	7.5.19
HIGH	harness_npm	brace-exp...@2.0.1	2.1.6 / 5.0.11

Supply-Chain & Container CVE Attribution (Shipped Browser Bundle vs. Dev Lockfile vs. Harness npm CLI)



AST Dangerous Code Sinks (CWE-79 / CWE-338) & Built-In Hypermedia Security Architecture

Detailed Security & Supply-Chain Audit Data

1. Supply-Chain & Trivy Vulnerability Breakdown

- **Production Browser Bundles (`dist/htmx.js`, `dist/htmx.esm.js`, `dist/htmax.js`): 0 dependencies, 0 CVEs**.
- **Secrets & Container IaC Hardening**: `0` hardcoded secrets (`SecurityFacts.secrets = []`, `AstFacts.hardcoded_secrets = []`, `committed_secret_files = []`) and `0` Dockerfile misconfigurations (`misconfigurations = []`).
- **Development Lockfile (`package-lock.json` `dev_or_optional` scope — `22` total CVEs: `0` CRITICAL, `5` HIGH, `15` MEDIUM, `2` LOW)**:
  - `[HIGH]` `CVE-2024-21538` in `cross-spawn@7.0.3` -> fix=`7.0.5, 6.0.6` (ReDoS in dev test runner subprocess spawning)
  - `[HIGH]` `CVE-2026-69192` in `ip-address@9.0.5` -> fix=`10.3.1` (IP address parsing inconsistency in dev server dependency)
  - `[HIGH]` `CVE-2026-102990` in `basic-ftp@5.3.1` -> fix=`6.2.1` (DoS via crafted directory listing in Playwright browser downloader dependency)
  - `[HIGH]` `CVE-2026-19693` & `CVE-2026-56876` in `extract-zip@2.0.1` (Symlink validation in Playwright zip extraction helper)

- `[MEDIUM]` `CVE-2026-82417` in `qs@6.15.3` -> fix=`6.16.0`
- `[MEDIUM]` `CVE-2026-104844` in `postcss-selector-parser@6.0.10` / `6.1.0` -> fix=`7.1.6, 6.1.5`

- **Test Harness Global `npm` CLI (`/usr/local/lib/node_modules/npm` inside non-shipped `node:22-bookworm` image — `29` findings)**:
  - Trivy's container image scan flagged `29` advisories in the base image's global `npm` CLI packages (`tar@6.2.1` `CVE-2026-59873` -> `7.5.19`, `brace-expansion@2.0.1` `CVE-2026-102276` -> `5.0.10`, `glob@10.4.2` `CVE-2025-64756` -> `10.5.0`, `minimatch@9.0.5` `CVE-2026-26996` -> `9.0.6`, `pacote@18.0.6` `CVE-2026-9496` -> `21.5.1`). These belong to the container's pre-installed `/usr/local/lib/node_modules/npm` binary and are never part of `htmx`.

2. Static CWE Sink Audit (`AstFacts.dangerous_sinks`, 8 Locations)

Severity	CWE ID	Location	Expression	Engineering Verdict & Remediation
`HIGH`	`CWE-79` (`xss`)	`src/ext/hx-head.js:59`	`htmlDoc.innerHTML = text`	Parses `<head>` markup from server responses; should wrap `text` via core's `#ttPolicy` helper for strict Trusted Types compatibility.
`HIGH`	`CWE-79` (`xss`)	`src/ext/hx-live.js:612`	`elt.innerHTML = val`	Reactive `q().html` setter; should route `val` through core's `#ttPolicy` wrapper (`P1-2`).
`HIGH`	`CWE-79` (`xss`)	`src/ext/hx-live.js:613`	`elt.outerHTML = val`	Reactive outerHTML setter; should route `val` through core's `#ttPolicy` wrapper (`P1-2`).
`HIGH`	`CWE-79` (`xss`)	`src/ext/hx-live.js:796`	`elt.innerHTML = ...`	Template initialization in `hx-live`; should route through `#ttPolicy` (`P1-2`).
`LOW`	`CWE-338`	`src/ext/hx-history-cache.js:26`	`Math.random()`	Generates non-security local cache key suffix; benign in browser context.
`LOW`	`CWE-338`	`src/ext/hx-multipart.js:18`	`Math.random()`	Generates internal multipart stream correlation ID; benign.
`LOW`	`CWE-338`	`src/ext/hx-sse.js:208`	`Math.random()`	Computes exponential backoff jitter for SSE reconnection; benign.
`LOW`	`CWE-338`	`src/ext/hx-ws.js:246`	`Math.random()`	Computes exponential backoff jitter for WebSocket reconnection; benign.

3. API Endpoint Attack Surface (`api_endpoints`)

- **Extracted Server HTTP Routes**: `0` (`htmx` is a client-side browser hypermedia library and CLI migration scanner; dynamic server penetration testing `SEC-06` is `not_applicable`).